Supplementary Material

Combustion chemistry of ammonia/hydrogen mixtures: Jet-stirred reactor measurements and comprehensive kinetic modeling

Xiaoyuan Zhang[a*], Shamjad P. Moosakutty[a], Rajitha P. Rajan[a], Mourad Younes[b], S. Mani Sarathy[a]

[a] *Clean Combustion Research Center, King Abdullah University of Science and Technology, Thuwal, 23955-6900, Saudi Arabia*

[b] *Saudi Aramco, Dhahran, 31311, Saudi Arabia*

S1 Additional validations of the present model

Table S1. A list of validation data for the present model.

	Data type	T(K)	P(atm)	ϕ	Reference
H_2/N_2O	Flame speed	300	0.8	0.8-2.2	[1]
	Ignition delay	956-1661	1-30	0.5	[2]
H_2/NO	JSR	700-1150	1-10	0.1-2.5	[3]
	Flow reactor	802	10	0.25	[4]
N_2H_4	Pyrolysis	1100-1190	2-3	∞	[5]
NH_3/NO	Flame speed	298	1	∞	[6]
	Flame speed	298	1	0.1-4.0	[7]
	JSR	1100-1450	1	0.1-2.0	[8]
	JSR	1000-1400	1	0.016	[9]
	Flow reactor	920-1380	1	0.002	[10]
	Flow reactor	940-1400	1	0.0016-0.08	[11]
	Burner-stabilized flame	1500-2100	0.071	1.46	[12]
NH_3/NO_2	Flow reactor	850-1350	1	∞	[13]
NH_3	Pyrolysis	2300-2800	1	∞	[14]
	Flame speed	298	1-5	0.6-1.5	[15]
	Flame speed	298	1	0.68-1.3	[16]
	Flame speed	298-473	1	0.8-1.4	[17]
	Flame speed	298-473	1-10	0.8-1.3	[18]
	Ignition delay	1560-2455	1.4-30	0.5-2.0	[19]
	Ignition delay	1100-1600	20-40	0.5-2.0	[20]
	JSR	1100-1450	1	0.1-2.0[a]	[8]
	JSR	500-1200	1.03	0.0093-0.0188	[21]
	Flow reactor	450-925	30-100	0.22-1.04	[22]
	Flow reactor	1100-2000	1.25	0.375	[21]
NH_3/H_2	Flame speed	298-473	1	0.8-1.4	[17]

* Corresponding author. E-mail: xiaoyuan.zhang@kaust.edu.sa.

Flame speed	298	1-5	1.0	[23]
Flame speed	473	1-10	0.8-1.4	[18]
Flame speed	298	1-5	0.7-1.6	[24]
Ignition delay	950-1150	20-60	0.5-2.0	[25]
Ignition delay	1020-1945	1.2-10	1.0	[26]
JSR	800-1280	1	0.25-1.0	Present work
Burner-stabilized flame	400-2000	0.05-0.12	0.9-1.1	[27]

[a]: This equivalence ratio is calculated based on the global reaction of $NH_3 + 1.25O_2 = NO + 1.5H_2O$ while the ϕ defined in other pure NH_3 experiments are based on $NH_3 + 0.75\ O_2 = 0.5N_2 + 1.5\ H_2O$.

S1.1. Oxidation of H_2/NO and H_2/N_2O

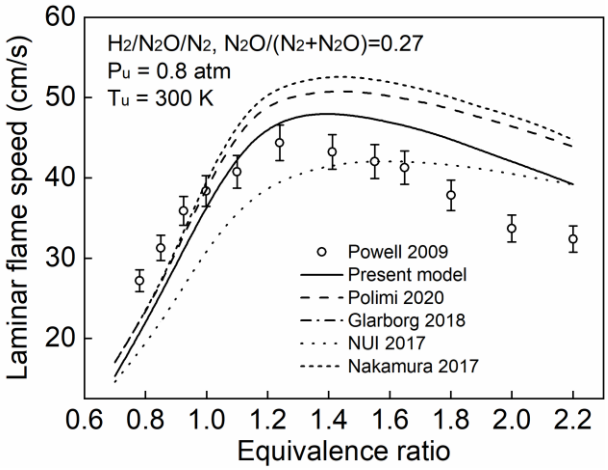


Figure S1. Measured (symbols) [1] and simulated (lines) laminar flame speeds of $H_2/N_2O/N_2$ mixtures at unburned temperature of 300 K and pressure of 0.8 atm. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

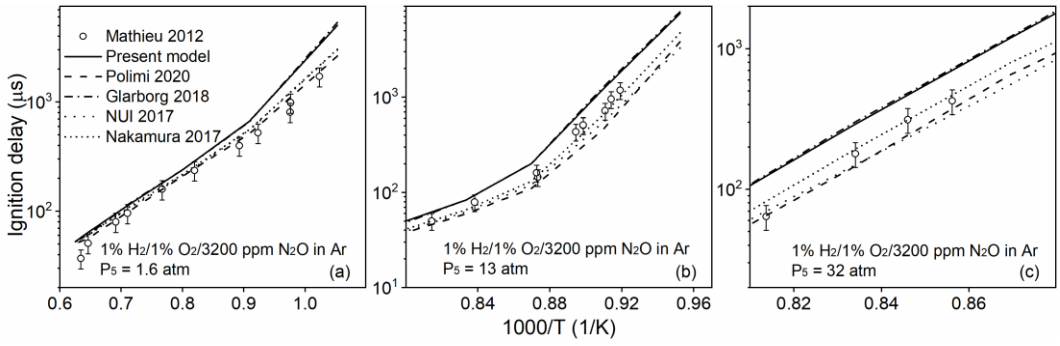


Figure S2. Measured (symbols) [2] and simulated (lines) ignition delay times of $H_2/N_2O/O_2/Ar$ mixtures at (a) 1.6 atm, (b) 13 atm and (c) 32 atm. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg

2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

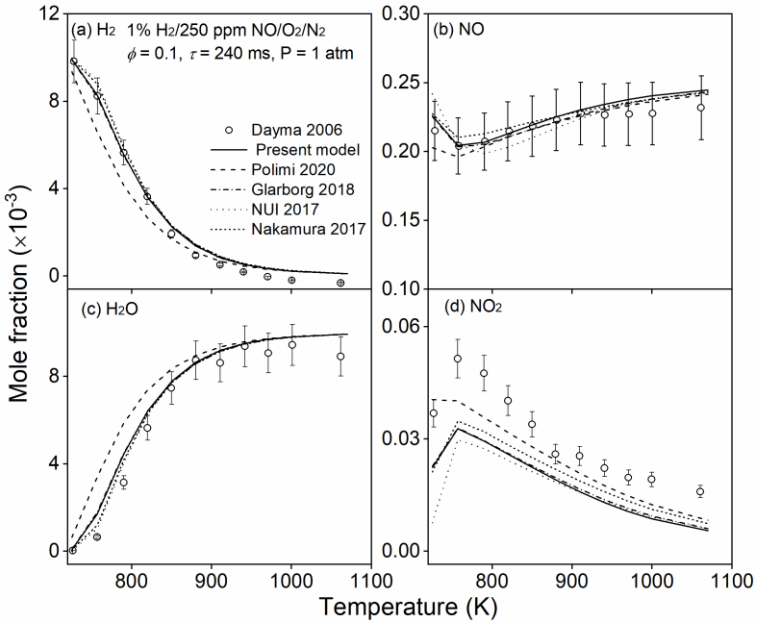


Figure S3. Mole fraction profiles of (a) H_2, (b) NO, (c) H_2O and (d) NO_2 in the JSR oxidation of H_2/NO mixture at 1 atm and $\phi = 0.1$. Symbols are the experimental data from Dayma et al. [3]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

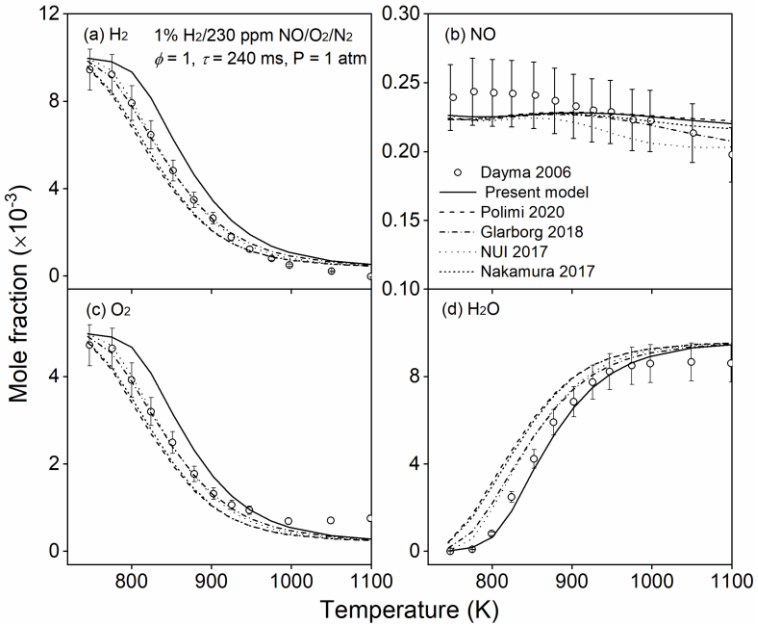


Figure S4. Mole fraction profiles of (a) H_2, (b) NO, (c) O_2 and (d) H_2O in the JSR oxidation of H_2/NO mixture at 1 atm and $\phi = 1.0$. Symbols are the experimental data from Dayma et al. [3]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated

results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

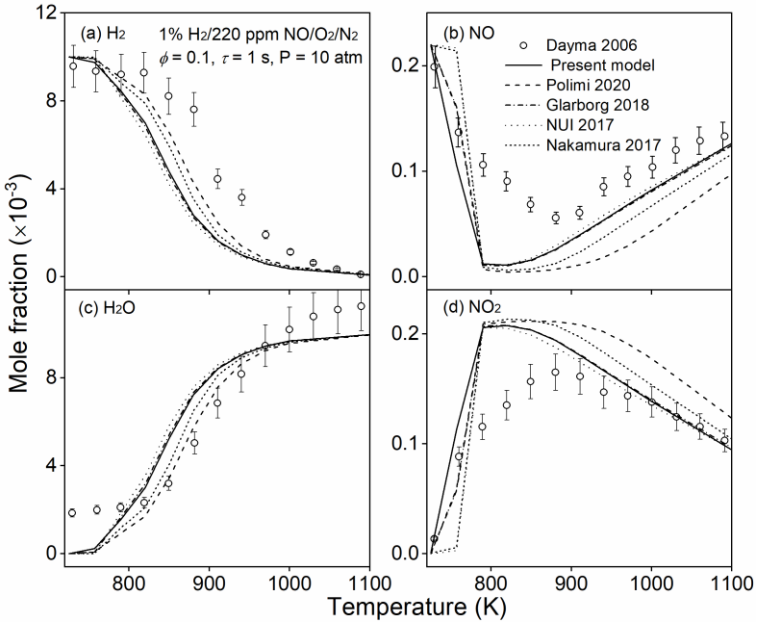


Figure S5. Mole fraction profiles of (a) H_2, (b) NO, (c) H_2O and (d) NO_2 in the JSR oxidation of H_2/NO mixture at 10 atm and $\phi = 0.1$. Symbols are the experimental data from Dayma et al. [3]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

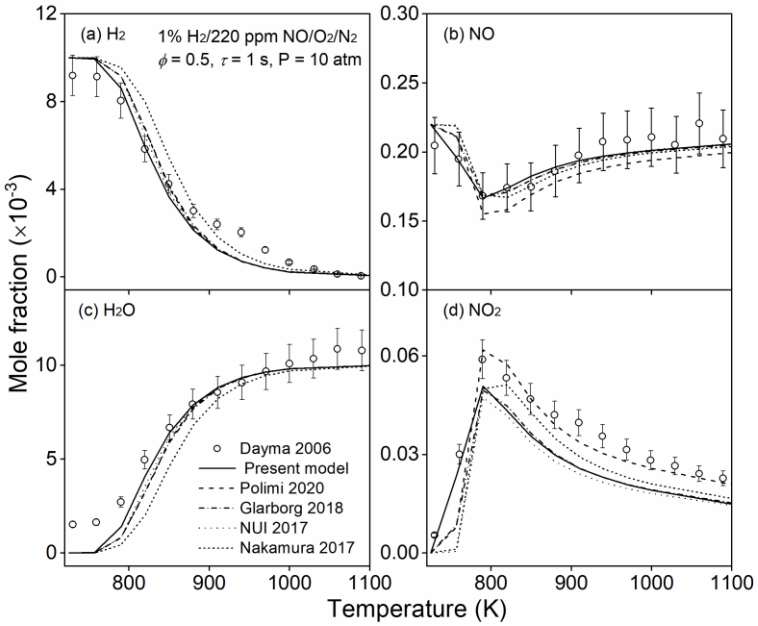


Figure S6. Mole fraction profiles of (a) H_2, (b) NO, (c) H_2O and (d) NO_2 in the JSR oxidation of H_2/NO mixture at 10 atm and $\phi = 0.5$. Symbols are the experimental data from Dayma et

al. [3]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

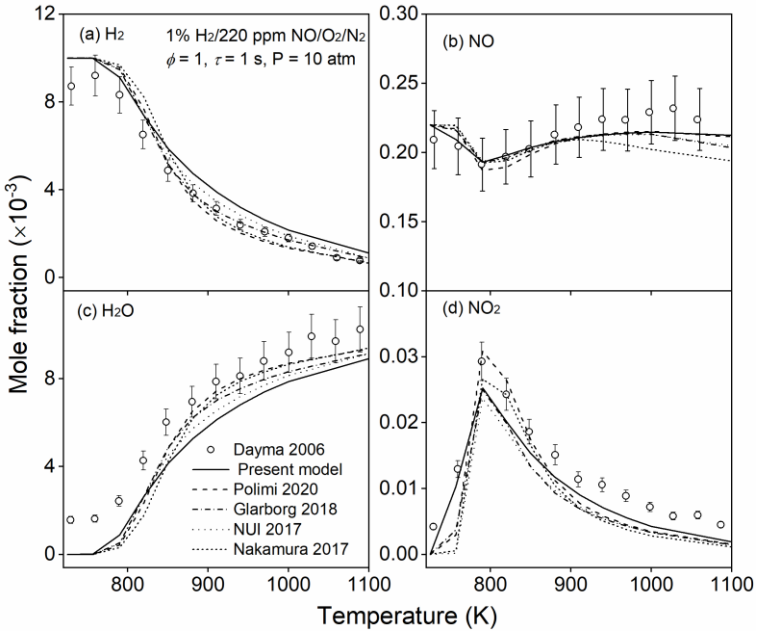


Figure S7. Mole fraction profiles of (a) H_2, (b) NO, (c) H_2O and (d) NO_2 in the JSR oxidation of H_2/NO mixture at 10 atm and ϕ = 1.0. Symbols are the experimental data from Dayma et al. [3]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

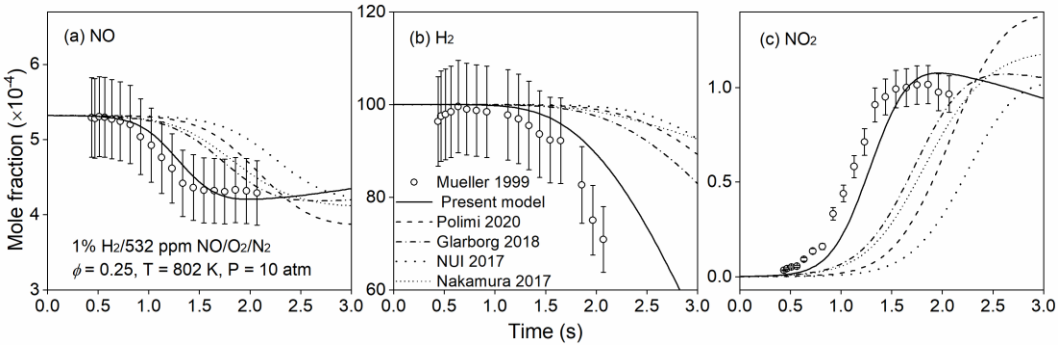


Figure S8. Mole fraction profiles of (a) NO, (b) H_2 and (c) NO_2 in the JSR oxidation of H_2/NO mixture at 802 K, 10 atm and ϕ = 25. Symbols are the experimental data from Mueller et al. [4]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and

Nakamura 2017 [30] models, respectively.

S1.2. Pyrolysis of NH_3 and N_2H_4

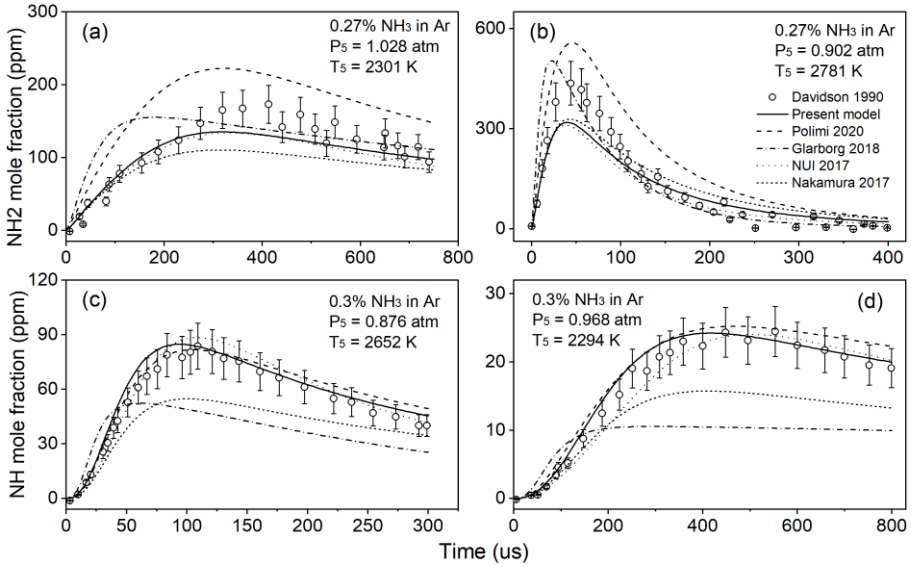


Figure S9. Mole fraction profiles of (a) NH_2, (b) NH_2, (c) NH and (d) NH in the shock tube pyrolysis of NH_3 under various conditions. Symbols are the experimental data from Davidson et al. [14]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

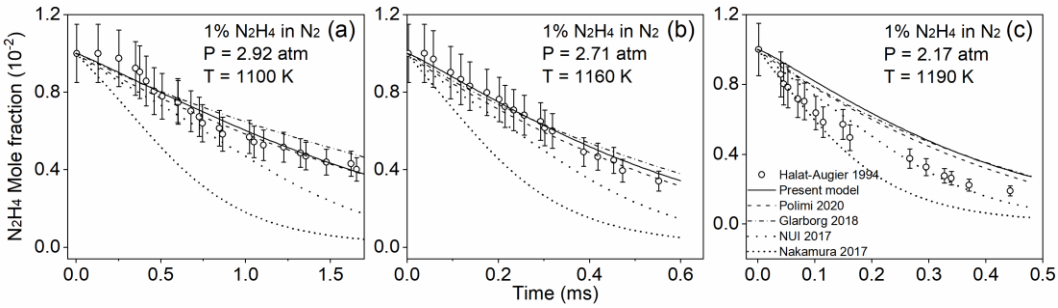


Figure S10. Mole fraction profiles of N_2H_4 in the pyrolysis of N_2H_4 at (a) 2.92 atm and 1100 K, (b) 2.71 atm and 1160 K and (c) 2.17 atm and 1190 K. Symbols are the experimental data from Halat-Augier [5]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

S1.3. Oxidation of NH_3/NO and NH_3/NO_2

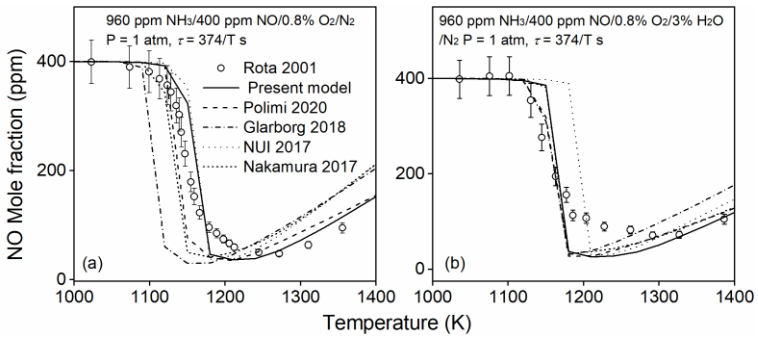


Figure S11. Mole fraction profiles of NO in the JSR oxidation of NH_3/NO with (a) 0% H_2O and (b) 3% H_2O at 1 atm and τ = 374/T s. Symbols are the experimental data from Rota et al. [9]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

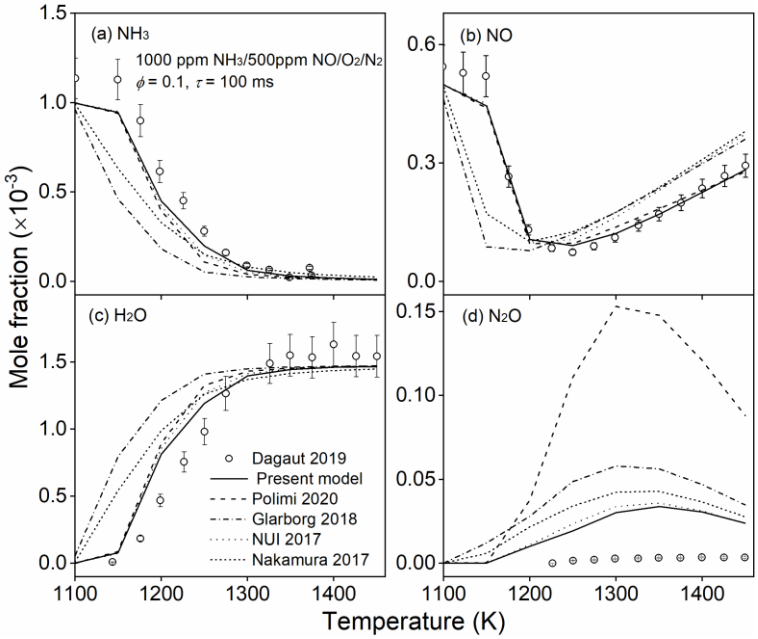


Figure S12. Mole fraction profiles of (a) NH_3, (b) NO, (c) H_2O and (d) N_2O in the JSR oxidation of NH_3/NO at 1 atm and ϕ = 0.1. Symbols are the experimental data from Daguat [8]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

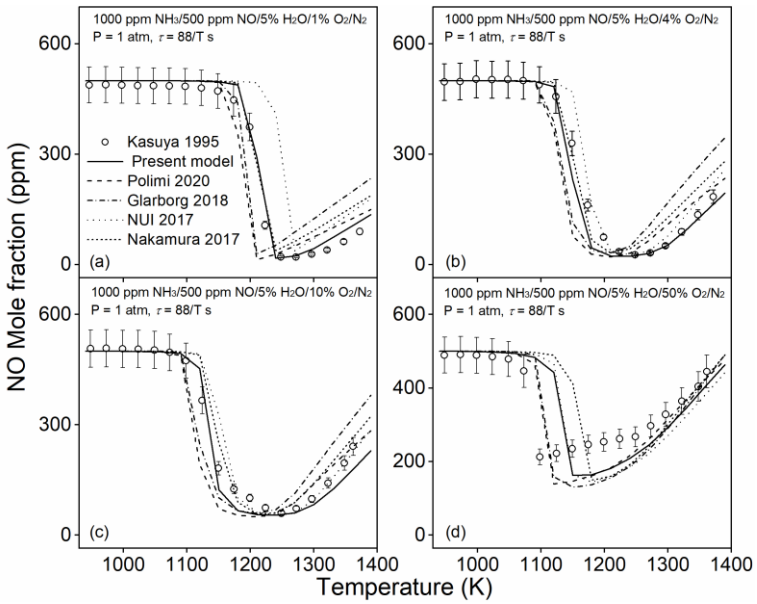


Figure S13. Mole fraction profiles of NO in the flow reactor oxidation of NH_3/NO with the O_2 content of (a) 1%, (b) 4%, (c) 10% and (d) 50% at 1 atm and τ = 88/T s. Symbols are the experimental data from Kasuya et al. [11]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

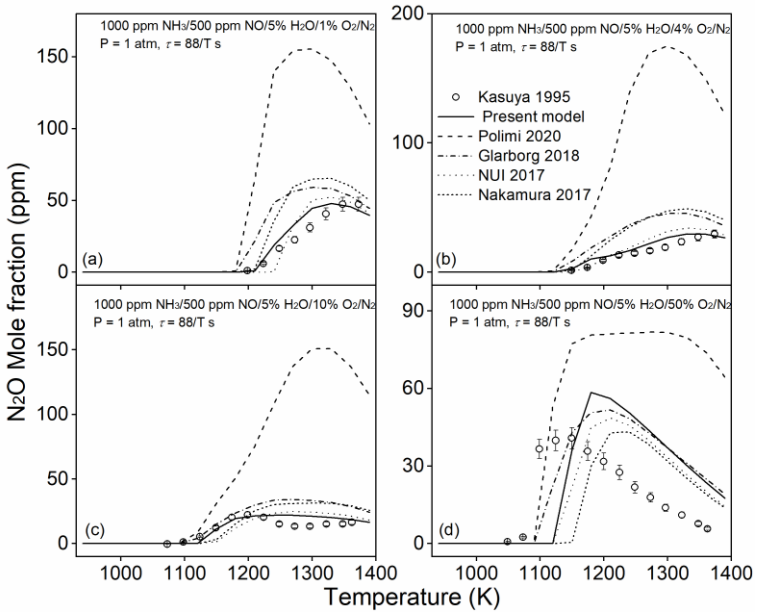


Figure S14. Mole fraction profiles of N_2O in the flow reactor oxidation of NH_3/NO with the O_2 content of (a) 1%, (b) 4%, (c) 10% and (d) 50% at 1 atm and τ = 88/T s. Symbols are the experimental data from Kasuya et al. [11]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg

2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

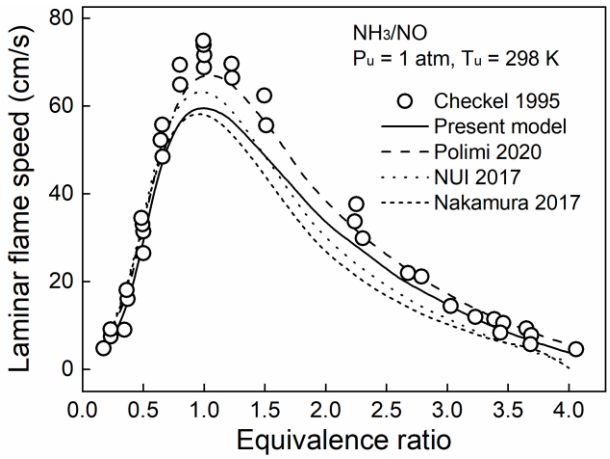


Figure S15. Laminar flame speed of NH_3/NO mixtures at unburned temperature of 298 K and pressure of 1 atm. Symbols are the experimental data from Checkel et al. [7]. Solid, dashed, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

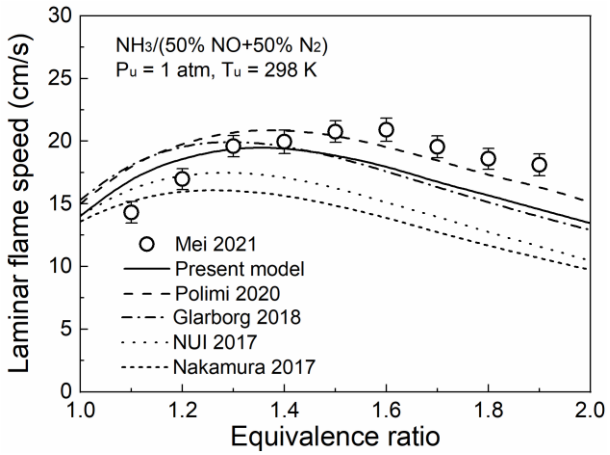


Figure S16. Laminar flame speed of NH_3/NO/N_2 mixtures at unburned temperature of 298 K and pressure of 1 atm. Symbols are the experimental data from Mei et al. [6]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

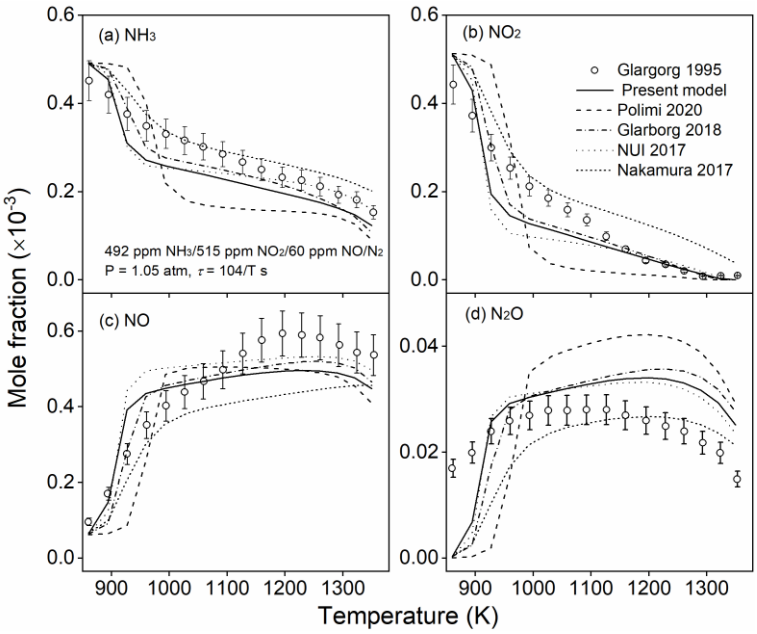


Figure S17. Mole fraction profiles of (a) NH_3, (b) NO_2, (c) NO and (d) N_2O in the flow reactor oxidation of 492 ppmNH_3/515 ppmNO_2/60 ppmNO/N_2 at 1.05 atm and τ = 104/T s. Symbols are the experimental data from Glarborg [13]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

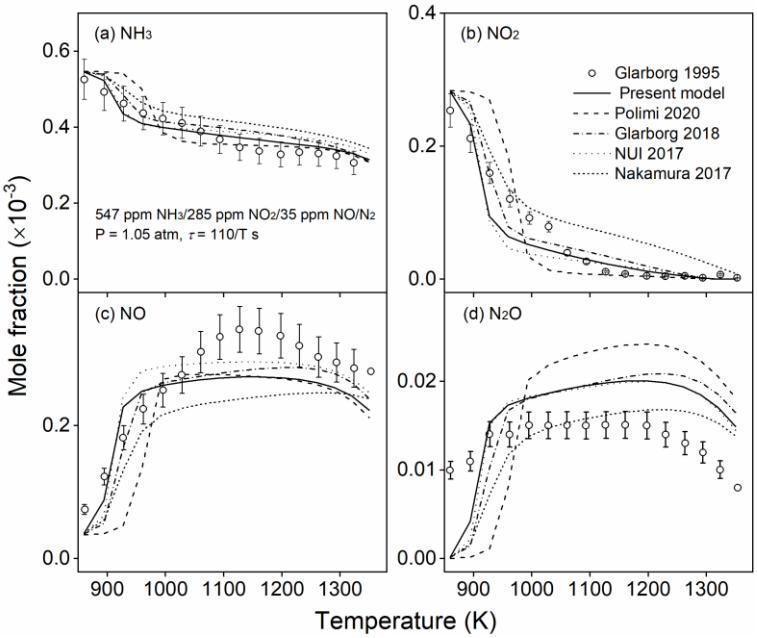


Figure S18. Mole fraction profiles of (a) NH_3, (b) NO_2, (c) NO and (d) N_2O in the flow reactor oxidation of 547 ppmNH_3/285 ppmNO_2/35 ppmNO/N_2 at 1.05 atm and τ = 110/T s.

Symbols are the experimental data from Glarborg et al. [13]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

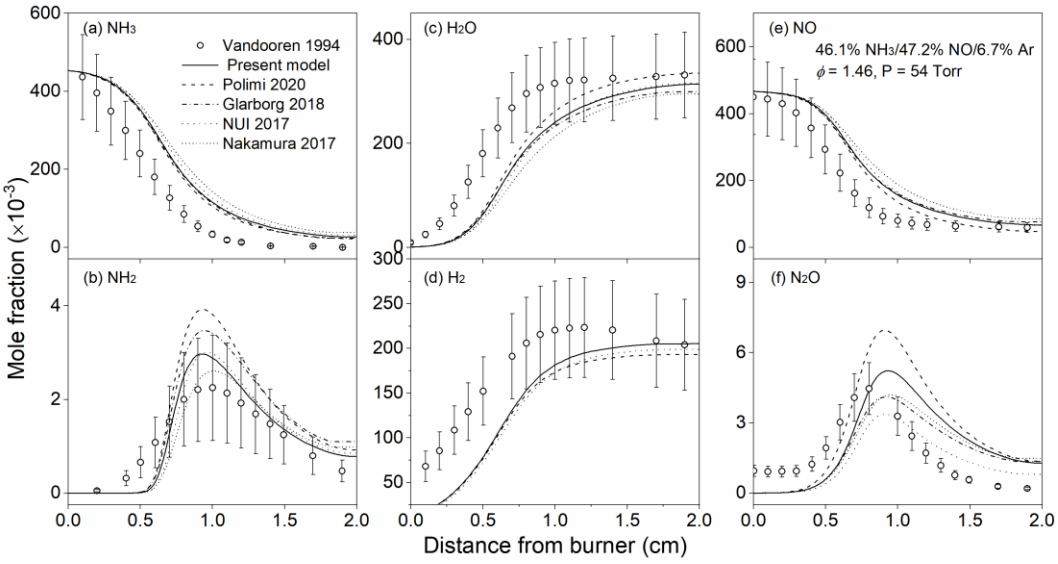


Figure S19. Mole fraction profiles of (a) NH_3, (b) NH_2, (c) H_2O, (d) H_2, (e) NO and (f) N_2O in the premixed flame of 46.1% NH_3/47.2% NO/6.7% Ar at 0.07 atm and ϕ = 1.46. Symbols are the experimental data from Vandooren et al. [12]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

S1.4. Oxidation of pure NH_3

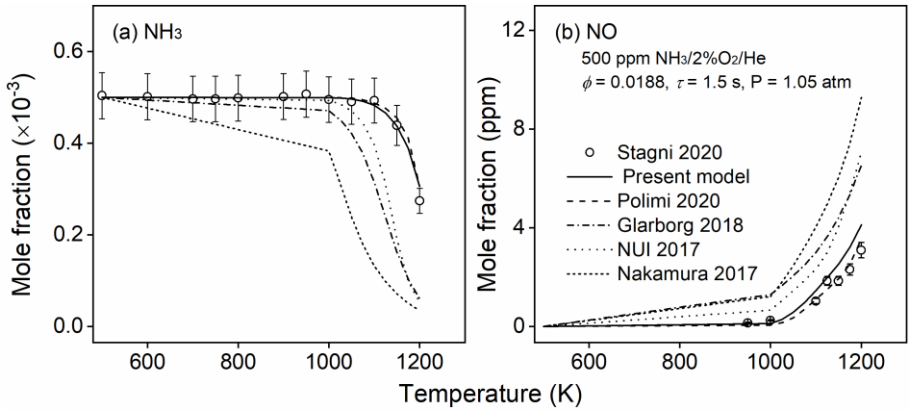


Figure S20. Mole fraction profiles of (a) NH_3 and (b) NO in the JSR oxidation of 500 ppm NH_3/2% O_2/He at 1.05 atm and ϕ = 0.0188. Symbols are the experimental data from Stagni et al. [21]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

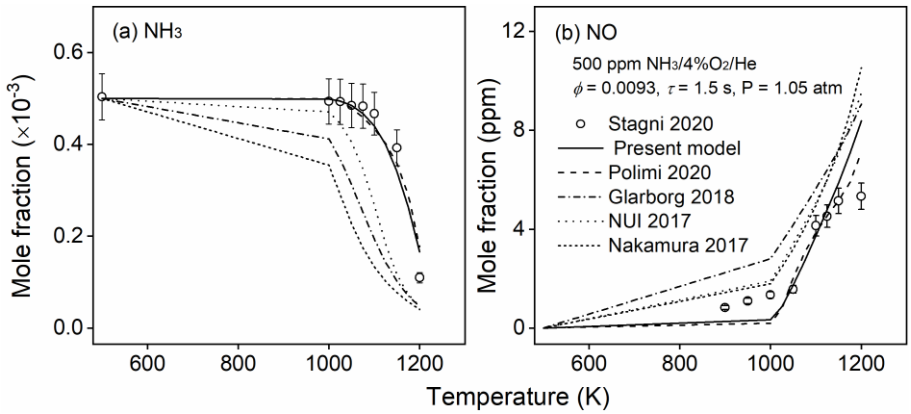


Figure S21. Mole fraction profiles of (a) NH_3 and (b) NO in the JSR oxidation of 500 ppm NH_3/4% O_2/He at 1.05 atm and ϕ = 0.0093. Symbols are the experimental data from Stagni et al. [21]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

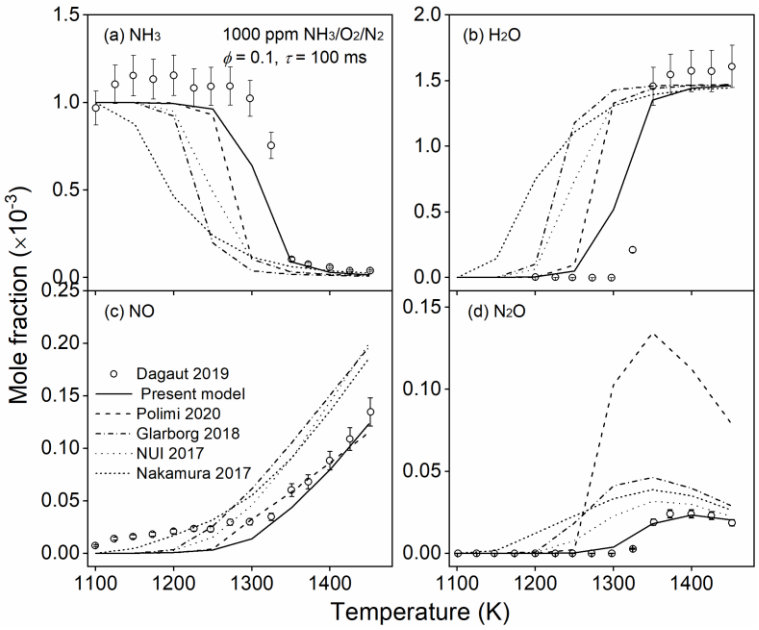


Figure S22. Mole fraction profiles of (a) NH_3, (b) H_2O, (c) NO and (d) N_2O in the JSR oxidation of $NH_3/O_2/N_2$ at 1 atm and ϕ = 0.1. Symbols are the experimental data from Dagaut [8]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

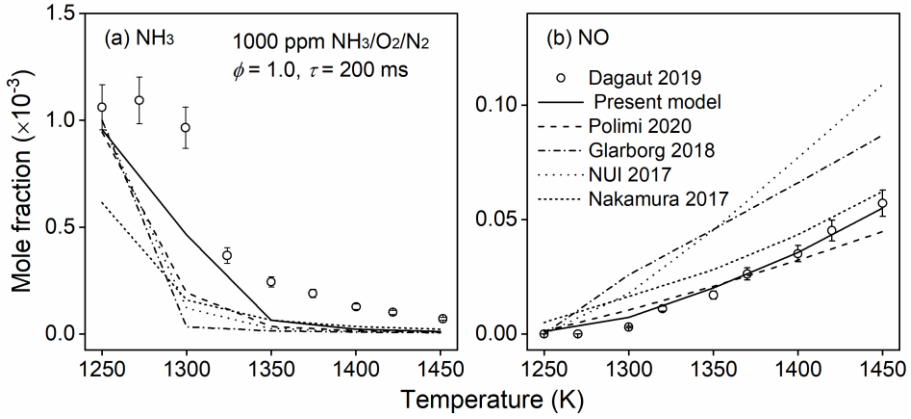


Figure S23. Mole fraction profiles of (a) NH_3 and (b) NO in the JSR oxidation of $NH_3/O_2/N_2$ at 1 atm and ϕ = 1.0. Symbols are the experimental data from Dagaut [8]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

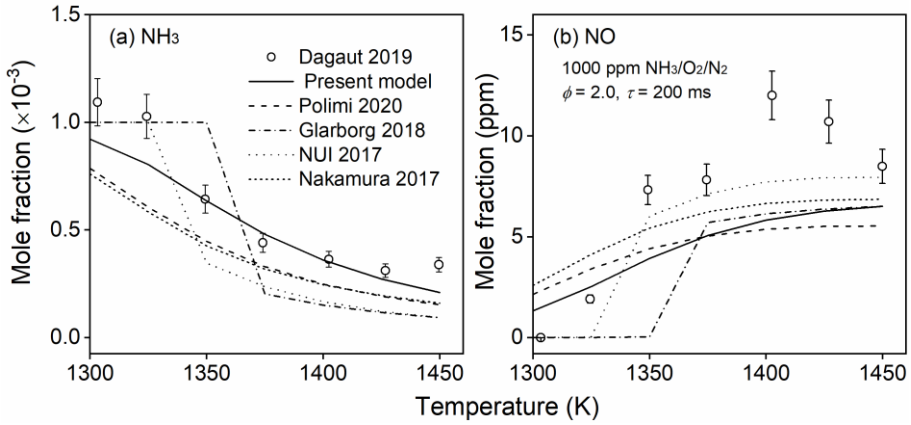


Figure S24. Mole fraction profiles of (a) NH_3 and (b) NO in the JSR oxidation of $NH_3/O_2/N_2$ at 1 atm and $\phi = 2.0$. Symbols are the experimental data from Dagaut [8]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

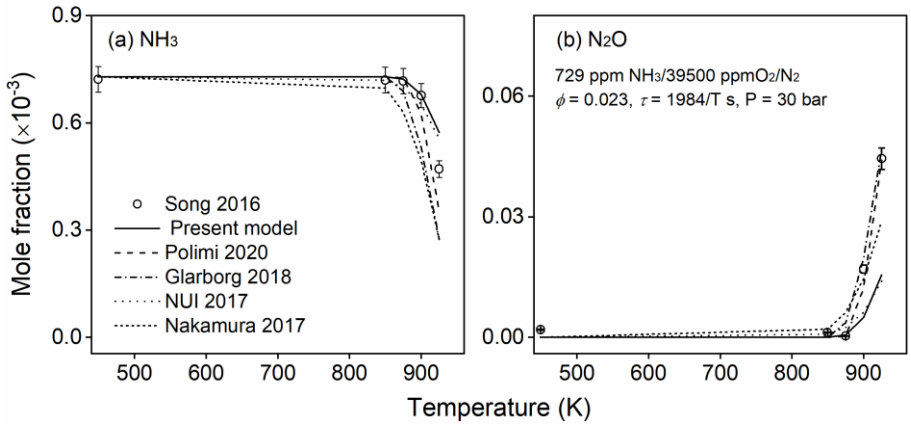


Figure S25. Mole fraction profiles of (a) NH_3 and (b) N_2O in the flow reactor oxidation of $NH_3/O_2/N_2$ at 30 atm and $\phi = 0.023$. Symbols are the experimental data from Song et al. [22]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

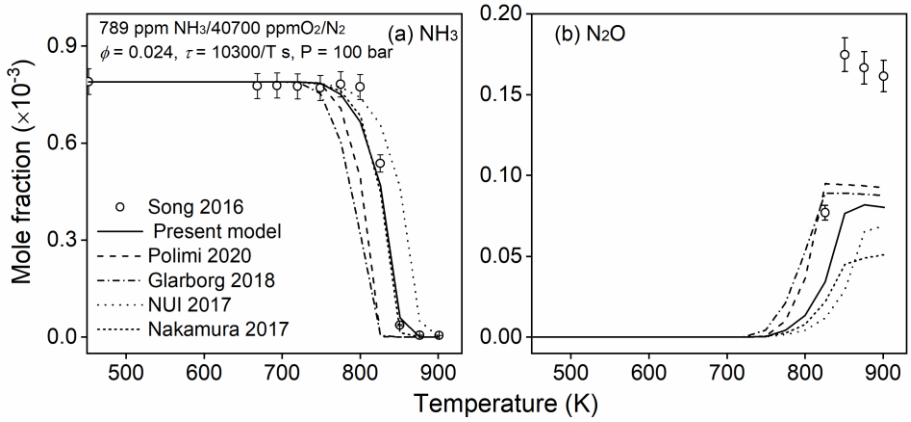


Figure S26. Mole fraction profiles of (a) NH_3 and (b) N_2O in the flow reactor oxidation of $NH_3/O_2/N_2$ at 100 atm and $\phi = 0.024$. Symbols are the experimental data from Song et al. [22]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

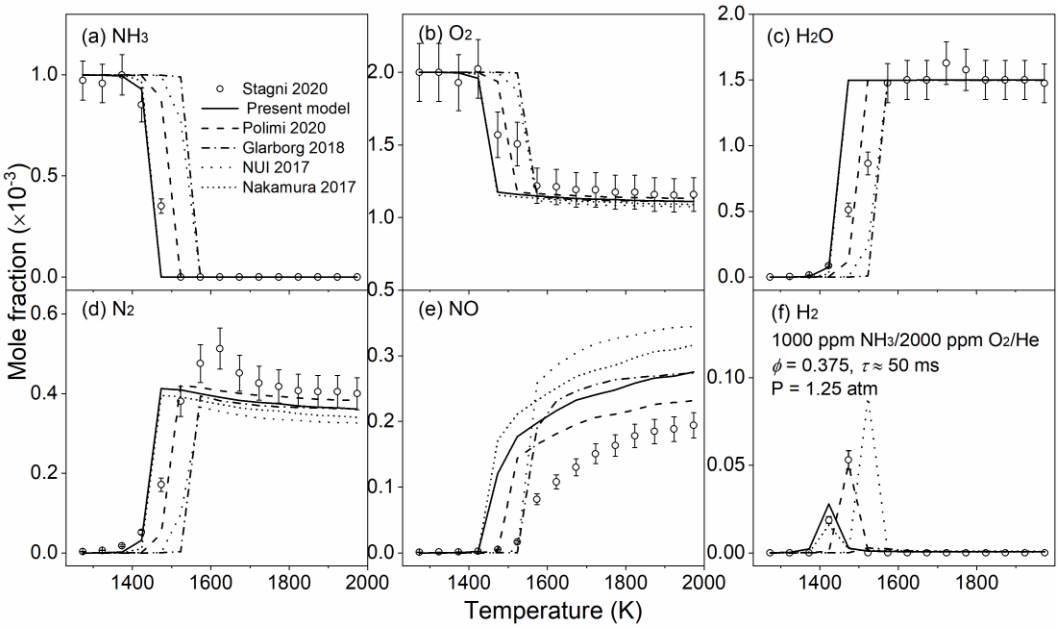


Figure S27. Mole fraction profiles of (a) NH_3, (b) O_2, (c) H_2O, (d) N_2, (e) NO and (f) H_2 in the flow reactor oxidation of NH_3/O_2/He at 1.25 atm and $\phi = 0.375$. Symbols are the experimental data from Stagni [21]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

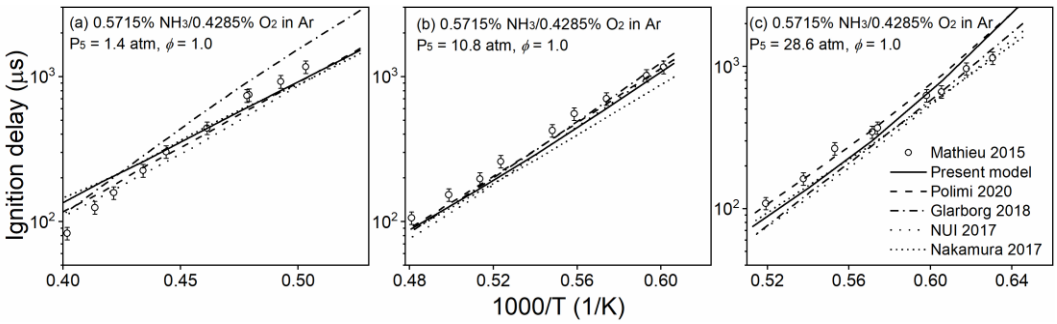


Figure S28. Ignition delay times of $NH_3/O_2/Ar$ mixture measured in a shock tube at $\phi = 1.0$ and (a) 1.4 atm, (b) 10.8 atm and (c) 28.6 atm. Symbols are the experimental data from Mathieu et al. [19]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

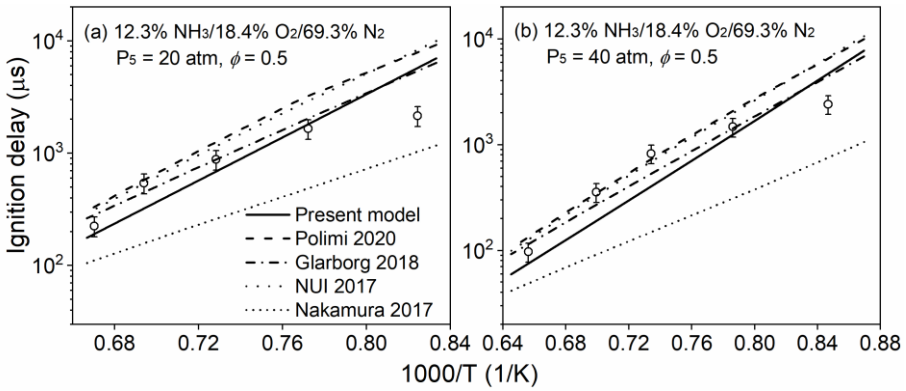


Figure S29. Ignition delay times of NH_3/air mixture measured in a shock tube at $\phi = 0.5$ and (a) 20 atm and (b) 40 atm. Symbols are the experimental data from Shu et al. [31]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

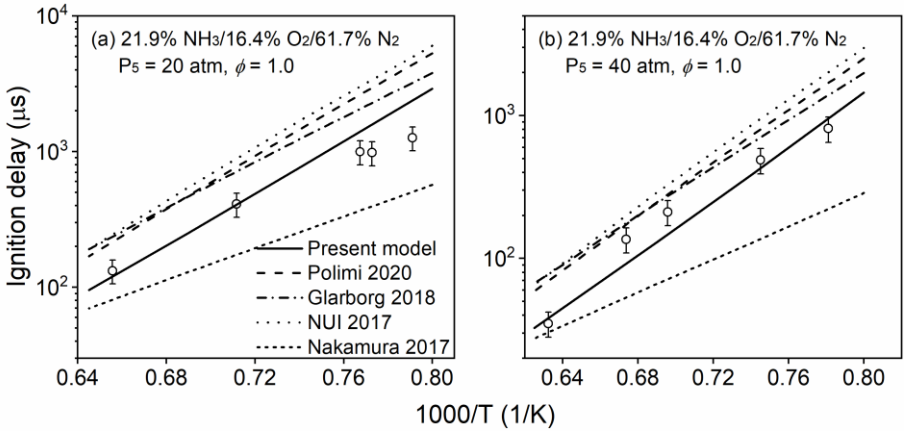


Figure S30. Ignition delay times of NH_3/air mixture measured in a shock tube at ϕ = 1.0 and (a) 20 atm and (b) 40 atm. Symbols are the experimental data from Shu et al. [31]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

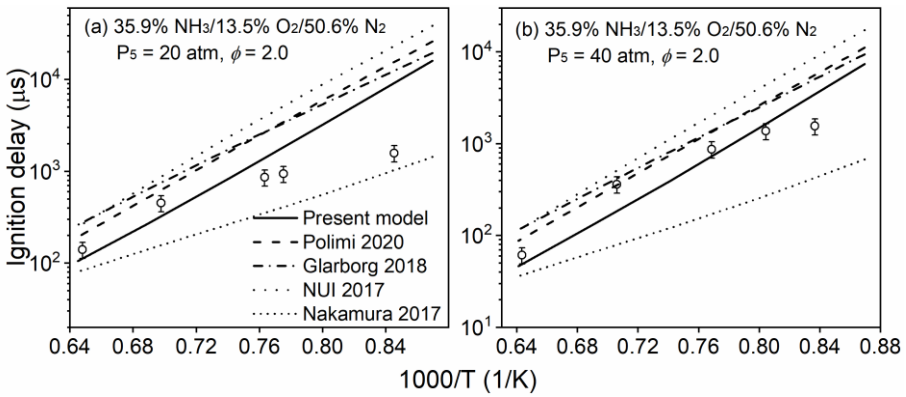


Figure S31. Ignition delay times of NH_3/air mixture measured in a shock tube at ϕ = 2.0 and (a) 20 atm and (b) 40 atm. Symbols are the experimental data from Shu et al. [31]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

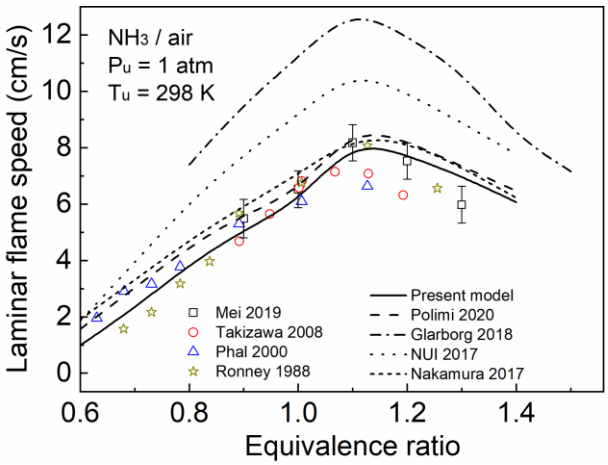


Figure S32. Laminar flame speeds of NH_3/air mixtures at an unburned temperature of 298 K and pressure of 1 atm. Symbols are the experimental data from[15,16,32,33]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

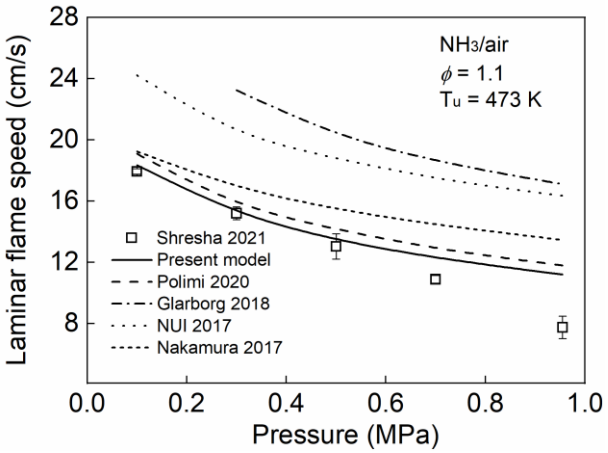


Figure S33. Laminar flame speeds of NH_3/air mixtures at an unburned temperature of 473 K and ϕ = 1.1. Symbols are the experimental data from Shresha et al. [18]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

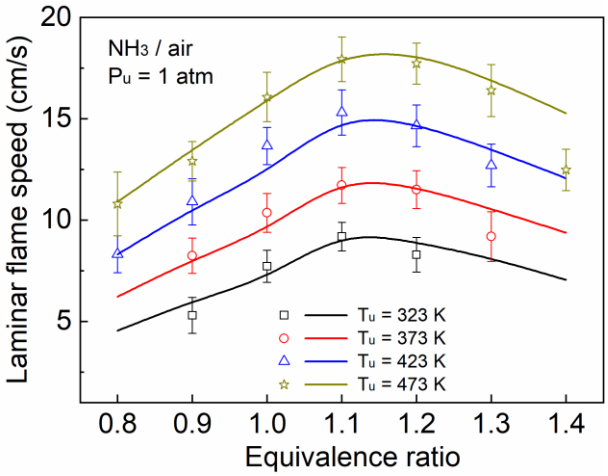


Figure S34. Laminar flame speeds of NH_3/air mixtures at an unburned pressure of 1 atm and temperatures of 323-473 K. Symbols are the experimental data from Lhuillier et al. [17]. Solid lines are the simulated results of the present model.

S1.5. Oxidation of NH_3/H_2 mixture

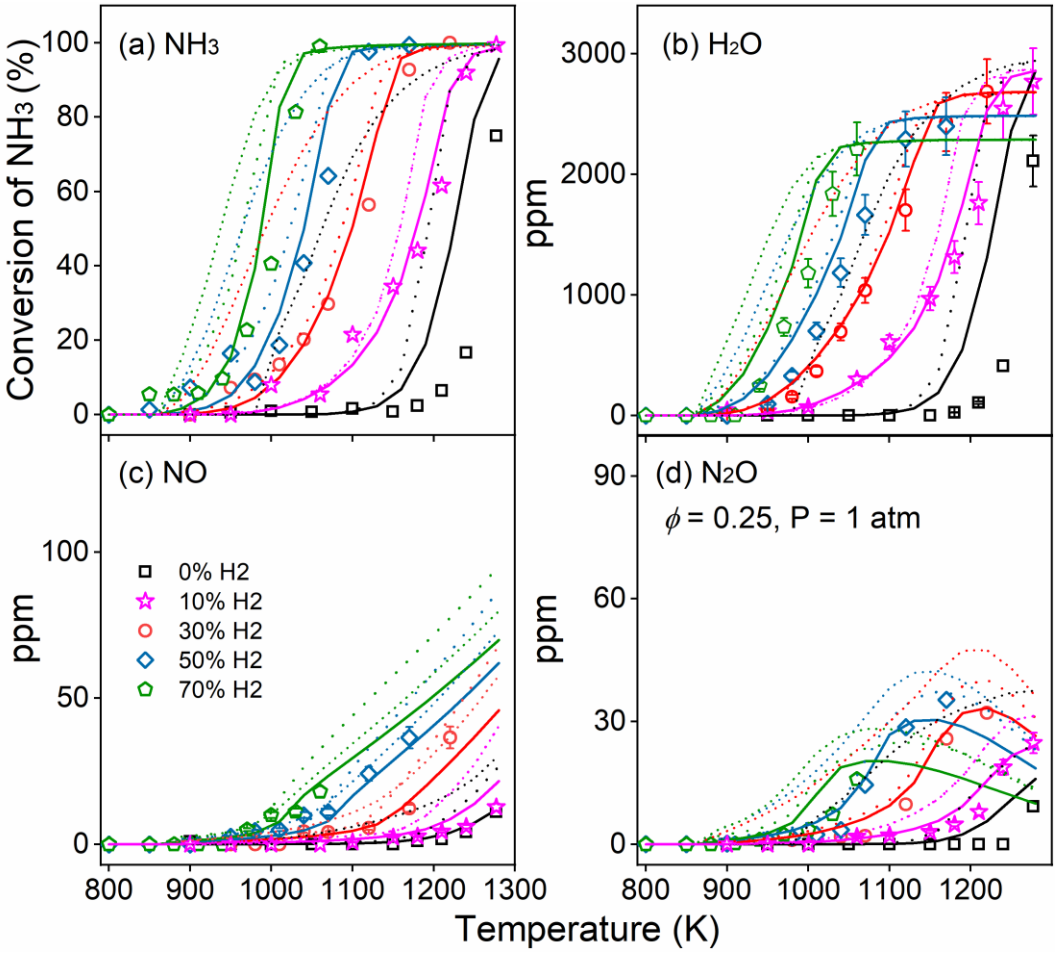


Fig. S35 Measured and simulated conversion ratio of (a) NH_3 and mole fractions of (b) H_2O, (c) NO and (d) N_2O in JSR oxidation of NH_3/H_2 mixtures with various H_2 content at $\phi = 0.25$, $\tau = 1$ s and P = 1 atm. Symbols are measured results from the present work. Lines are simulated results by different models. Solid, sparse dotted and dense dotted lines are the present, NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

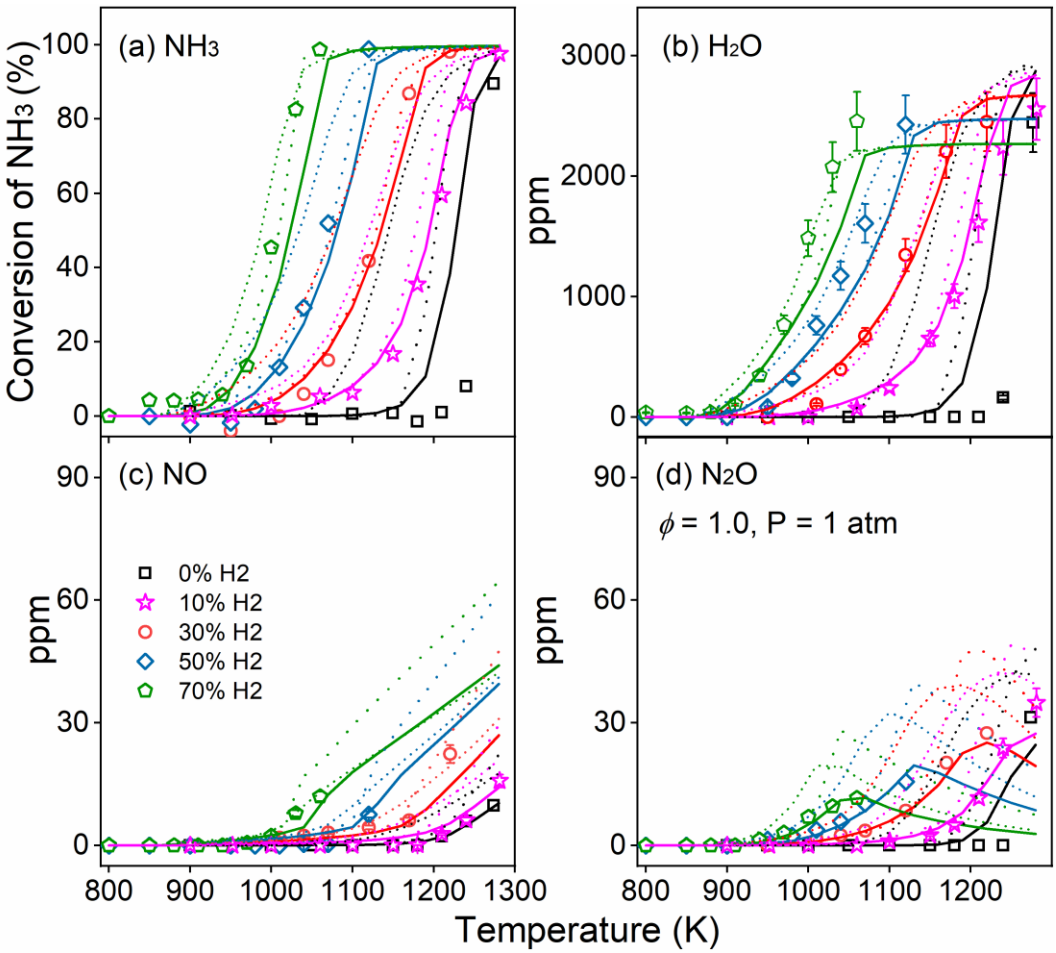


Fig. S36 Measured and simulated conversion ratio of (a) NH_3 and mole fractions of (b) H_2O, (c) NO and (d) N_2O in JSR oxidation of NH_3/H_2 mixtures with various H_2 content at $\phi = 1$, $\tau = 1$ s and P = 1 atm. Symbols are measured results from the present work. Lines are simulated results by different models. Solid, sparse dotted and dense dotted lines are the present, NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

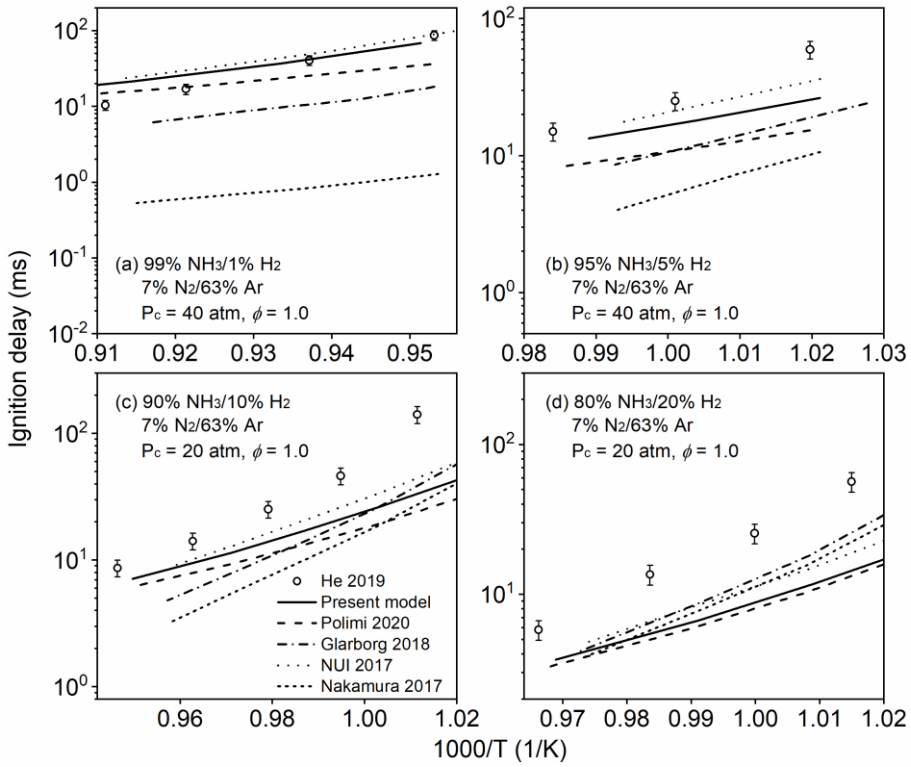


Figure S37. Ignition delay times of $NH_3/H_2/O_2/N_2/Ar$ mixtures measured in an RCM at ϕ = 1.0. (a) 99% NH_3/1% H_2, (b) 95% NH_3/5% H_2, (c) 90% NH_3/10% H_2 and (d) 80% NH_3/20%H_2. Symbols are the experimental data from He et al. [25]. Solid, dashed, dash dot, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], Glarborg 2018 [28], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

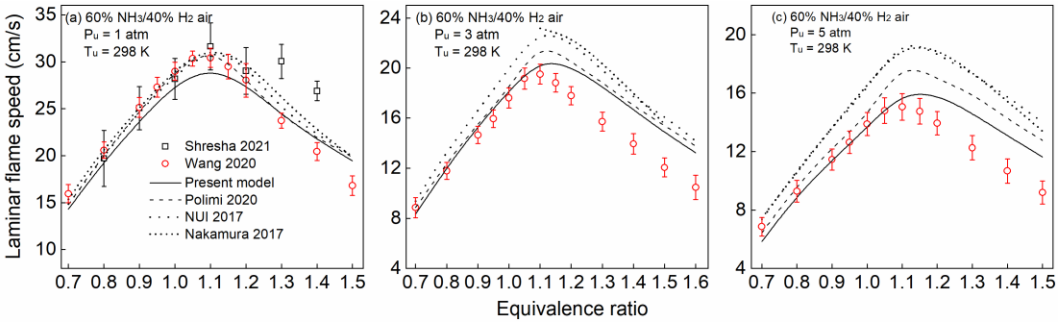


Figure S38. Laminar flame speeds of 60% NH_3/40% H_2/air mixtures at an unburned temperature of 298 K and (a) P_u = 1 atm, (b) P_u = 3 atm and (c) P_u = 5 atm. Squares and circles are the experimental data from Shresha et al. [18] and Wang et al. [24], respectively. Solid, dashed, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

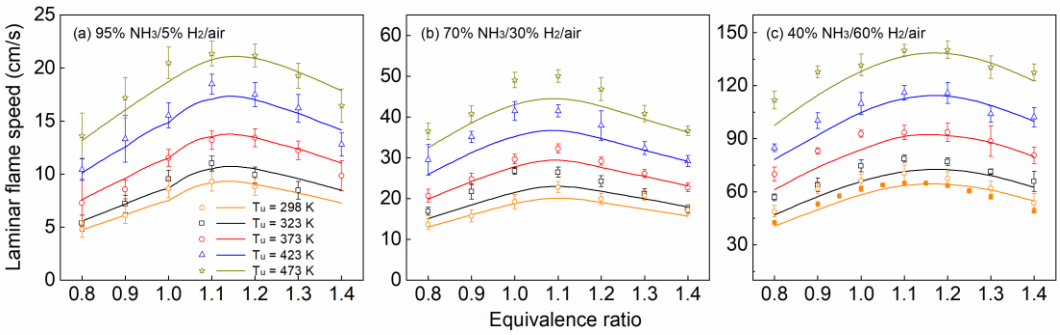


Figure S39. Laminar flame speeds of (a) 95% NH_3/5% H_2/air mixtures, (b) 70% NH_3/30% H_2/air mixtures and (c) 40% NH_3/60% H_2/air mixtures at an unburned temperature of 298-473 K and P_u = 1 atm. Symbols are the experimental data from Lhuillier et al. [17]. Solid lines are the simulated results of the present model.

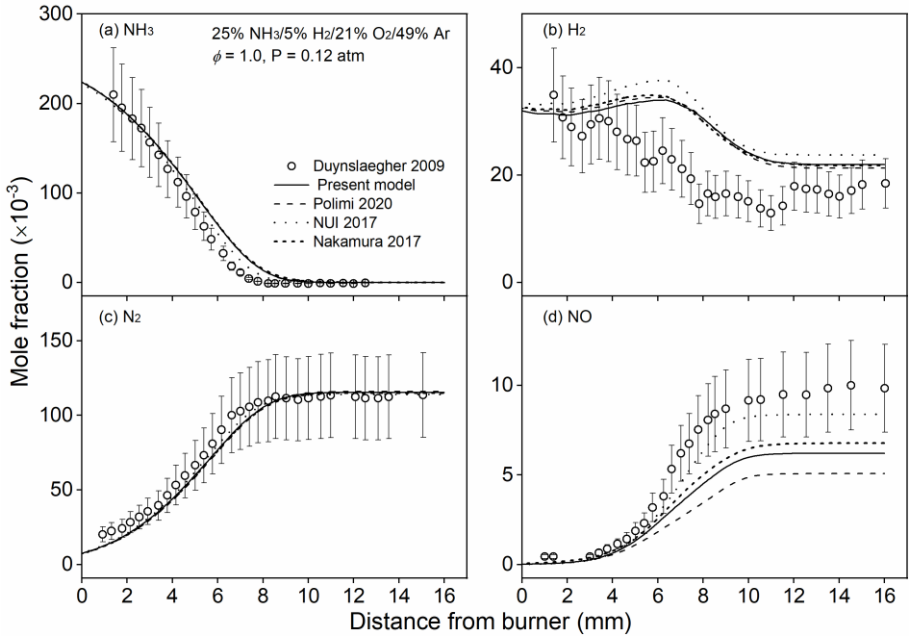


Figure S40. Species profiles of (a) NH_3, (b) H_2, (c) N_2 and (d) NO in a laminar premixed flame of NH_3/H_2/O_2/Ar mixture at pressure of 0.12 atm and ϕ = 1.0. Symbols are the experimental data from Duynslaegher et al. [27]. Solid, dashed, sparse dotted and dense dotted lines are the simulated results of the present, Polimi 2020 [21], NUI 2017 [29] and Nakamura 2017 [30] models, respectively.

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